Greatest Coefficient and Greatest Term

1. Let $(2+3x)^{12} = \sum_{k=0}^{12} t_k x^k$, where $t_k = {}^{12}C_k \times 2^{12-k} \times 3^k$ is the coefficient of x^k. Write down the greatest coefficient (and leave it factored).
2. Let $(7+3x)^{25} = \sum_{k=0}^{25} c_k x^k$. Write down the greatest coefficient (and leave it factored)
3. Let $(3+4x)^{13} = \sum_{k=0}^{13} T_k$, where $T_k = {}^{13}C_k \times 3^{13-k} \times (4x)^k$ is the term in x^k. Write down the greatest term when $x = \frac{1}{2}$ (and leave it factored).
4. Let $(1+5x)^{21} = \sum_{k=0}^{21} T_k$, where T_k is the term in x^k. Write down the greatest term when $x = \frac{3}{5}$ (and leave it factored).
5. a. Let $(5+2x)^{15} = \sum_{k=0}^{15} t_k x^k$. Write down the greatest coefficient (and leave it factored).
   b. Let $(5+2x)^{15} = \sum_{k=0}^{15} T_k$, where T_k is the term in x_k. Write down the greatest term when $x = \frac{5}{3}$ (and leave it factored).
6. For each expansion find: (i) the greatest coefficient, (ii) the greatest term if $x = \frac{2}{3}$.
   a. $(1+4x)^{11}$
   b. $\left(2+\frac{3}{4}x\right)^9$
   c. $(5x+3)^{12}$
   d. $(5+6x)^{11}$
7. For each of the following expansions, find: (i) the coefficient with greatest absolute value, (ii) the term with greatest absolute value if $x = \frac{2}{3}$ and $y = 3$.
   a. $(1-7x)^9$
   b. $(7-2x)^{14}$
   c. $(x-2y)^{12}$
   d. $(2x-y)^{15}$
8. For each of the following expansions, find: (i) the greatest coefficient, (ii) the greatest term if $x = \frac{3}{2}$ and $y = \frac{4}{9}$.
   a. $\left(2x^2+\frac{3}{x}\right)^{10}$
   b. $(2x+3y)^{12}$
9. Show that in the expansion of $(1+x)^{14}$, where $x = \frac{2}{3}$, two consecutive terms are equal to each other and greater than any other term.
10. Let $(x+y)^n = \sum_{r=0}^{n} T_r$, where T_r is the term in $x^{n-r}y^r$.
    a. Write down expressions for T_r and T_{r+1} and whence show that $\frac{T_{r+1}}{T_r} = \frac{(n-r)y}{(r+1)x}$.
    b. Consider the expansion of $(a+3b)^8$, where $a = 2$ and $b = \frac{3}{4}$. By substituting the appropriate values for x, y and n into the expansion in (i), show that $\frac{T_{r+1}}{T_r} = \frac{72-9r}{8r+8}$. Hence show that T_4 is the numerically greatest term in the expansion.
    c. Find the greatest term in the expansion $(p+q)^{10}$, where $p = q = \frac{1}{2}$.
11. Let $(1+x)^n = \sum_{r=0}^{n} t_r x^r$.
    a. Find the values of n and r if $t_{r+1} = 5t_r$ and $t_{r+4} = 2t_{r+3}$.
    b. Hence find the greatest coefficient in the expansion.
12. Let $(1+0.01)^{12} = \sum_{r=0}^{12} T_r$, where T_r is the term in $(0.01)^r$. Find the first value of r for which the ratio $\frac{T_{r+1}}{T_r} < 0.005$.
13. Let $(\sin\theta + \cos\theta)^{20} = \sum_{k=0}^{20} T_k$, where T_k is the term in $\sin^{20-k}\theta\cos^k\theta$. Find, to the nearest degree, the first positive angle for which $T_{14} > T_{15}$.
14. a. Show that in the expansion of $(1+x)^n$, where n is an even integer, the term with the greatest coefficient is the term in $x^{\frac{1}{2}n}$.
    b. By considering the expansion of $(1+x)^{2n}$, for any positive integer n, prove that the largest value of ${}^{2n}C_r$ is ${}^{2n}C_n$.

Answer

1. ${}^{12}C_7 2^5 3^7$
2. ${}^{25}C_7 7^{18} 3^7$
3. ${}^{13}C_5 3^8 2^5$
4. ${}^{21}C_{16} 3^{16}$
5. a. ${}^{15}C_4 5^{11} 2^4$; b. ${}^{15}C_6 5^{15} \left(\frac{2}{3}\right)^6$
6. a. i. ${}^{11}C_9 4^9$, ii. $T_8 = {}^{11}C_8 \left(\frac{8}{3}\right)^8$; b. i. ${}^{9}C_2 2^3 3^2$, ii. $T_1 = T_2 = 1152$; c. i. ${}^{12}C_8 3^4 5^8$, ii. $T_6 = {}^{12}C_6 10^6$; d. i. ${}^{11}C_6 5^5 6^6$, ii. $T_5 = {}^{11}C_5 5^6 4^5$
7. a. i. ${}^{9}C_8 7^8$, ii. $T_8 = {}^{9}C_8 \left(\frac{14}{3}\right)^8$; b. i. $-{}^{14}C_3 7^{11} 2^3$, ii. $T_2 = {}^{14}C_2 7^{12} \left(\frac{4}{3}\right)^2$; c. i. ${}^{12}C_8 2^8$, ii. $T_{11} = -{}^{15}C_{11} 4^4 3^7$
8. a. i. ${}^{10}C_6 2^4 3^6$, ii. ${}^{10}C_3 9^7 2^{-4}$; b. i. ${}^{12}C_7 2^5 3^7$, ii. Two terms have the greatest valie, which is 10,264,320
9. The equal terms are ${}^{14}C_6 \left(\frac{2}{3}\right)^6$ and ${}^{14}C_5 \left(\frac{2}{3}\right)^5$.
10. a. $T_{r+1} = {}^{n}C_{r+1} x^{n-r-1} y^{r+1}$, $T_r = {}^{n}C_r x^{n-r} y^r$; c. $T_5 = \frac{63}{256}$
11. a. $n = 17, r = 2$; b. ${}^{17}C_8 = {}^{17}C_9$
12. $r = 8$
13. $\theta = 22°$